

LAPORAN
PRAKTIKUM SISTEM MULTIMEDIA

Tugas 1

“Setup Tools Praktikum & Representasi Digital Multimedia”

Dosen Pengampu : Dewi Tresnawati, S.Pt., M.T.



Disusun oleh :

Muhammad Taufiq

2406074

Informatika C

FAKULTAS ILMU KOMPUTER
PROGRAM STUDI TEKNIK INFORMATIKA
INSTITUT TEKNOLOGI GARUT
2026

1.1. Tujuan Praktikum (sesuai tujuan pada modul pertemuan)

1. Mahasiswa mampu menginstal dan mengonfigurasi Python, pip, FFmpeg, Audacity,

Jupyter Notebook/VS Code, Pillow, dan NumPy pada komputer praktikum.

2. Mahasiswa mampu membaca metadata berbagai format file multimedia menggunakan FFprobe.

3. Mahasiswa mampu melakukan konversi format dasar menggunakan FFmpeg.

1.2. Alat dan Bahan

- Python 3.x
- pip
- FFmpeg & FFprobe
- Audacity
- Pillow (PIL)
- NumPy
- Jupyter Notebook / VS Code
- Koneksi internet (unduh installer)

1.3. Langkah Kerja

1. Instalasi Python dan pengelola paket

1. Unduh dan instal Python 3.x dari python.org, pastikan opsi "Add to PATH" dicentang saat instalasi (Windows).

2. Verifikasi instalasi: jalankan perintah ``python --version`` dan ``pip --version`` pada terminal.

3. Instal pustaka praktikum: ``pip install pillow numpy scipy opencv-python scikit-image PyWavelets jupyter``.

2. Instalasi FFmpeg dan FFprobe

- 1. Unduh FFmpeg build sesuai sistem operasi dari situs resmi ffmpeg.org.**
- 2. Tambahkan folder `bin` FFmpeg ke environment variable PATH.**
- 3. Verifikasi dengan menjalankan `ffmpeg -version` dan `ffprobe -version` pada terminal.**

3. Instalasi Audacity dan editor kerja

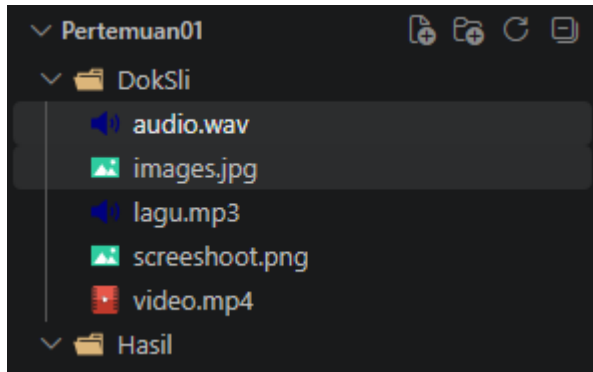
- 1. Unduh dan instal Audacity dari audacityteam.org.**
- 2. Instal Visual Studio Code atau gunakan Jupyter Notebook sebagai lingkungan kerja praktikum.**

4. Eksplorasi metadata multimedia

- 1. Siapkan 5 file dengan format berbeda, misalnya: .jpg, .png, .mp3, .wav, .mp4.**
- 2. Jalankan `ffprobe -hide\_banner nama\_file` untuk masing-masing file dan catat informasi codec, bitrate, durasi, dan resolusi/sample rate.**
- 3. Lakukan satu kali konversi format dasar, misalnya `ffmpeg -i input.wav output.mp3`, lalu bandingkan metadata sebelum dan sesudah konversi.**

Note: Saya sudah install semua software dan library, jadi hanya ada bukti instalasinya saja (Screenshoot terminal / GUI). Buktinya ada di Lampiran...

1.4. Hasil dan Analisis



Disini saya sudah menambahkan 5 file dengan format berbeda.

1. Informasi ke 5 file sebelum dikonversi

```
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner DokSli/audio.wav
Input #0, wav, from 'DokSli/audio.wav':
  Metadata:
    encoder      : Lavf63.1.101
  Duration: 00:02:27.89, bitrate: 1411 kb/s
  Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / 0x0001), 44100 Hz, 2 channels, s16, 1411 kb/s

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner DokSli/images.jpg
Input #0, image2, from 'DokSli/images.jpg':
  Duration: 00:00:00.04, start: 0.000000, bitrate: 4160 kb/s
  Stream #0:0: Video: mjpeg (Baseline), yuvj444p(pc, bt470bg/unknown/unknown), 554x554 [SAR 1:1 DAR 1:1], 25 fps, 25 tbr, 25 tbn

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner DokSli/lagu.mp3
Input #0, mp3, from 'DokSli/lagu.mp3':
  Metadata:
    title       : I'll leave my heart with your air
    artist      : So let me fly with you
    :           :
    :           : #yungkai #blue
    url         : https://www.youtube.com/watch?v=98zHKN-xSHk
    comment     : https://www.youtube.com/watch?v=98zHKN-xSHk
    date        : 2024
  Duration: 00:03:34.36, start: 0.023021, bitrate: 366 kb/s
  Stream #0:0: Audio: mp3 (mp3float), 48000 Hz, stereo, fltp, 320 kb/s, start 0.023021
  Metadata:
    encoder      : Lavc58.13
  Stream #0:1: Video: png, rgb24(pc, gbr/unknown/unknown), 1280x720, 90k tbr, 90k tbn, start 0.023022 (attached pic)
  Metadata:
    title        : "Album cover"
    comment      : Cover (front)

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner DokSli/screeshoot.png
Input #0, png_pipe, from 'DokSli/screeshoot.png':
  Duration: N/A, bitrate: N/A
  Stream #0:0: Video: png, rgba(pc, gbr/bt709/iec61966-2-1), 1919x1070 [SAR 3779:3779 DAR 1919:1070], 25 fps, 25 tbr, 25 tbn

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner DokSli/video.mp4
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'DokSli/video.mp4':
  Metadata:
    major_brand   : mp42
    minor_version : 0
    compatible_brands: mp42mp41isomavc1
    creation_time  : 2021-07-22T11:55:15.000000Z
  Duration: 00:00:22.24, start: 0.000000, bitrate: 24956 kb/s
  Stream #0:0[0x1](und): Video: h264 (High) (avc1 / 0x31637661), yuv420p(tv, bt709, progressive), 2160x3840, 24953 kb/s, 25 fps, 25 tbr, 25 tbn (default)
  Metadata:
    creation_time  : 2021-07-22T11:55:15.000000Z
    handler_name    : L-SMASH Video Handler
    encoder         : AVC Coding
```

2. Ketika proses konversi file

```

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffmpeg -i DokS11/audio.wav Hasil/audio.mp3
ffmpeg version 9.0.2-essentials_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
  configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --enable-fontconfig --enable-iconv --enable-gnutls --en
  able-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zlib --enable-lisrt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-l
  ibx264 --enable-libx265 --enable-libxvid --enable-libaom --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-libass --enable-libfreetype --enable-libfridi
  --enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg --enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --ena
  ble-ffnvcodec --enable-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-opengl --enable-libgme --enable-libopenmpt --enable-libopencore-amrwb --enable-libmp3lame --
  enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
  libavutil 61. 1.102 / 61. 1.102
  libavcodec 63. 1.102 / 63. 1.102
  libavformat 63. 1.102 / 63. 1.102
  libavdevice 63. 1.102 / 63. 1.102
  libavfilter 12. 1.102 / 12. 1.102
  libswscale 10. 1.102 / 10. 1.102
  libswresample 7. 1.102 / 7. 1.102
[aist#0:0/pcm_s16le @ 000002a535fbad0] Guessed Channel Layout: stereo
Input #0, wav, from 'DokS11/audio.wav':
  Metadata:
    encoder      : Lavf63.1.101
    Duration: 00:02:27.89, bitrate: 1411 kb/s
    Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / ex0001), 44100 Hz, stereo, s16, 1411 kb/s
  Stream mapping:
    Stream #0:0 -> #0:0 (pcm_s16le (native) -> mp3 (libmp3lame))
  Press [q] to stop, [?] for help
Output #0, mp3, to 'Hasil/audio.mp3':
  Metadata:
    TSSE        : Lavf63.1.102
    Stream #0:0: Audio: mp3, 44100 Hz, stereo, s16p
  Metadata:
    encoder      : Lavc63.1.102 libmp3lame
[Out#0/mp3 @ 000002a535fb1800] video:0KiB audio:231KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.010647%
size= 2312KiB time=00:02:27.88 bitrate= 128.1kbits/s speed=53.7x elapsed=0:00:02.75
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffmpeg -i DokS11/images.jpg Hasil/images.jpeg
ffmpeg version 9.0.2-essentials_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
  configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --enable-fontconfig --enable-iconv --enable-gnutls --en
  able-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zlib --enable-lisrt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-l
  ibx264 --enable-libx265 --enable-libxvid --enable-libaom --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-libass --enable-libfreetype --enable-libfridi
  --enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg --enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --ena
  ble-ffnvcodec --enable-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-opengl --enable-libgme --enable-libopenmpt --enable-libopencore-amrwb --enable-libmp3lame --
  enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
  libavutil 61. 1.102 / 61. 1.102
  libavcodec 63. 1.102 / 63. 1.102
  libavformat 63. 1.102 / 63. 1.102
  libavdevice 63. 1.102 / 63. 1.102
  libavfilter 12. 1.102 / 12. 1.102
  libswscale 10. 1.102 / 10. 1.102
  libswresample 7. 1.102 / 7. 1.102
Input #0, image2, from 'DokS11/images.jpg':
  Duration: 00:00:00.04, start: 0.000000, bitrate: 4160 kb/s
  Stream #0:0: Video: mjpeg (Baseline), yuvj444p(pc, bt470bg/unknown/unknown), 554x554 [SAR 1:1 DAR 1:1], 25 fps, 25 tbr, 25 tbn
  Stream mapping:
    Stream #0:0 -> #0:0 (mjpeg (native) -> mjpeg (native))
  Press [q] to stop, [?] for help
Output #0, image2, to 'Hasil/images.jpeg':
  Metadata:
    encoder      : Lavf63.1.102
    Stream #0:0: Video: mjpeg, yuvj444p(pc, bt470bg/unknown/unknown, progressive), 554x554 [SAR 1:1 DAR 1:1], q=2-31, 200 kb/s, 25 fps, 25 tbn
  Metadata:
    encoder      : Lavc63.1.102 mjpeg
  Side data:
    CPB properties: bitrate max/min/avg: 0/0/200000 buffer size: 0 vbv_delay: N/A
[Image2 @ 00000238e2ba3000] The specified filename 'Hasil/images.jpeg' does not contain an image sequence pattern or a pattern is invalid.
[Image2 @ 00000238e2ba3000] Use a pattern such as %03d for an image sequence or use the -update option (with -frames:v 1 if needed) to write a single image.
[Out#0/image2 @ 00000238e2b92000] video:19KiB audio:0KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: unknown
frame= 1 fps=0.0 q=5.6 lsize=N/A time=00:00:00.04 bitrate=N/A speed=1.17x elapsed=0:00:00.03

```

```

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffmpeg -i DokSli\lagu.mp3 Hasil\lagu.wav
ffmpeg version 9.0.2-essentials_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
configuration: --enable-gpl --enable-version3 --enable-static --disable-x32threads --disable-autodetect --enable-cairo --enable-fontconfig --enable-iconv --enable-gnutls --en
able-libaom2 --enable-gmp --enable-lzma --enable-lzss --enable-zlib --enable-zlib-ng --enable-libbrotli --enable-libdav1d --enable-libfreetype --enable-libgsm --enable-libj
ib264 --enable-libx265 --enable-libxvid --enable-libaom --enable-libopenjpeg --enable-libvpx --enable-libmediacodec --enable-libass --enable-libfribidi --enable-libharfbuzz
--enable-libtheora --enable-libvorbis --enable-libvpx --enable-libzmq --enable-libzstd --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb --enable-libopus --enable-lspspeex --enable-libvorbis --enable-librubberband
libavutil 61. 1.102 / 61. 1.102
libavcodec 63. 1.102 / 63. 1.102
libavformat 63. 1.102 / 63. 1.102
libavdevice 63. 1.102 / 63. 1.102
libavfilter 12. 1.102 / 12. 1.102
libswscale 10. 1.102 / 10. 1.102
libswresample 7. 1.102 / 7. 1.102
Input #0, mp3, from 'DokSli\lagu.mp3':
Metadata:
  artist       : yung kai
  title        : yung kai - blue (Official Audio)
  encoder      : Lavf58.76.100
  description   : Stream my track and follow me on socials here: https://linktr.ee/yungkaiboy
  :
  : lyrics:
  : [Verse 1]
  : Your morning eyes, I could stare like watching stars
  : I could walk you by, and I'll tell without a thought
  : You'd be mine, would you mind if I took your hand tonight?
  : Know you're all that I want this life
  :
  : [Chorus]
  : I'll imagine we fell in love
  : I'll nap under moonlight skies with you
  : I think I'll picture us, you with the waves
  : The oceans colors on your face
  : I'll leave my heart with your air
  : So let me fly with you
  : Will you be forever with me?
  :
  : [Verse 2]
  : My love will always stay by you
  : I'll keep it safe so don't you worry a thing, I'll tell you I love you more
  : It's stuck with you forever so promise you won't let it go
  : I'll trust the universe will always bring me to you
  :
  : [Chorus]
  : I'll imagine we fell in love
  : I'll nap under moonlight skies with you
  : I think I'll picture us, you with the waves
  : The oceans colors on your face
  : I'll leave my heart with your air
  : So let me fly with you
  :
  : #yungkai #blue
  : Stream my track and follow me on socials here: https://linktr.ee/yungkaiboy
  :
synopsis
:
: #yungkai #blue
: Stream my track and follow me on socials here: https://linktr.ee/yungkaiboy
:
: lyrics:
: [Verse 1]
: Your morning eyes, I could stare like watching stars
: I could walk you by, and I'll tell without a thought
: You'd be mine, would you mind if I took your hand tonight?
: Know you're all that I want this life
:
: [Chorus]
: I'll imagine we fell in love
: I'll nap under moonlight skies with you
: I think I'll picture us, you with the waves
: The oceans colors on your face
: I'll leave my heart with your air
: So let me fly with you
: Will you be forever with me?
:
: [Verse 2]
: My love will always stay by you
: I'll keep it safe so don't you worry a thing, I'll tell you I love you more
: It's stuck with you forever so promise you won't let it go
: I'll trust the universe will always bring me to you
:
: [Chorus]
: I'll imagine we fell in love
: I'll nap under moonlight skies with you
: I think I'll picture us, you with the waves
: The oceans colors on your face
: I'll leave my heart with your air
: So let me fly with you
:
: #yungkai #blue
: https://www.youtube.com/watch?v=98zHKN-xSHk
ICMT : https://www.youtube.com/watch?v=98zHKN-xSHk
ISFT : Lavf63.1.102
Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / 0x0001), 48000 Hz, stereo, s16, 1536 kb/s
Metadata:
  encoder      : Lavc63.1.102 pcm_s16le
[out#0/wav @ 00000211ebf43bc0] video:0KiB audio:40192KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.000496%
size= 40192KiB time=00:03:34.35 bitrate=1536.0kb/s speed= 585x elapsed=0:00:00.36

```

```

PS C:\Kulliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffmpeg -i DokSli\screshoot.png Hasil\screshoot.jpg
ffmpeg version 9.0.2-essentials build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
  configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --enable-fontconfig --enable-iconv --enable-gnutls --enable-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zlib --enable-lbsrt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-l
libx264 --enable-libx265 --enable-libxvid --enable-libvpx --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-libass --enable-libfreetype --enable-libfridi
--enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg --enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --enable-ffnvcodec --enable-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-opengl --enable-libgme --enable-libopenmpt --enable-libopencore-amrwb --enable-libmp3lame --enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
  libavutil      61. 1.102 / 61. 1.102
  libavcodec     63. 1.102 / 63. 1.102
  libavformat    63. 1.102 / 63. 1.102
  libavdevice    63. 1.102 / 63. 1.102
  libavfilter    12. 1.102 / 12. 1.102
  libswscale     10. 1.102 / 10. 1.102
  libswresample   7. 1.102 / 7. 1.102
Input #0, png_pipe, from 'DokSli\screshoot.png':
  Duration: N/A, bitrate: N/A
  Stream #0:0: Video: png, rgba(pc, gbr/bt709/iec61966-2-1), 1919x1070 [SAR 3779:3779 DAR 1919:1070], 25 fps, 25 tbr, 25 tbn
Stream mapping:
  Stream #0:0 -> #0:0 (png (native) -> mjpeg (native))
Press [q] to stop, [?] for help
Output #0, image2, to 'Hasil\screshoot.jpg':
  Metadata:
    encoder      : Lavf63.1.102
  Stream #0:0: Video: mjpeg, yuv444p(pc, unknown/bt709/iec61966-2-1, progressive), 1919x1070 [SAR 1:1 DAR 1919:1070], q-2-31, 200 kb/s, 25 fps, 25 tbn
  Metadata:
    encoder      : Lavc63.1.102 mjpeg
  Side data:
    CPB properties: bitrate max/min/avg: 0/0/200000 buffer size: 0 vbv_delay: N/A
[image2 @ 000001e0000c180] The specified filename 'Hasil\screshoot.jpg' does not contain an image sequence pattern or a pattern is invalid.
[image2 @ 000001e0000c180] Use a pattern such as %03d for an image sequence or use the -update option (with -frames:v 1 if needed) to write a single image.
[out0/image2 @ 000001e0000a140] video:121KiB audio:0KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: unknown
frames= 1 fps=0.0 q=7.7 Lsize=N/A time=00:00:00.04 bitrate=N/A speed=0.404x elapsed=0:00:00.09

```

```

PS C:\Kulliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffmpeg -i DokSli\video.mp4 Hasil\video.mkv
ffmpeg version 9.0.2-essentials build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
  configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --enable-fontconfig --enable-iconv --enable-gnutls --enable-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zlib --enable-lbsrt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-l
libx264 --enable-libx265 --enable-libxvid --enable-libvpx --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-libass --enable-libfreetype --enable-libfridi
--enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg --enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --enable-ffnvcodec --enable-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-opengl --enable-libgme --enable-libopenmpt --enable-libopencore-amrwb --enable-libmp3lame --enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
  libavutil      61. 1.102 / 61. 1.102
  libavcodec     63. 1.102 / 63. 1.102
  libavformat    63. 1.102 / 63. 1.102
  libavdevice    63. 1.102 / 63. 1.102
  libavfilter    12. 1.102 / 12. 1.102
  libswscale     10. 1.102 / 10. 1.102
  libswresample   7. 1.102 / 7. 1.102
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'DokSli\video.mp4':
  Metadata:
    major_brand   : mp42
    minor_version : 0
    compatible_brands: mp42mp4isomavc1
    creation_time  : 2021-07-22T11:55:15.000000Z
  Duration: 00:00:22.24, start: 0.000000, bitrate: 24956 kb/s
  Stream #0:0[0x1](und): Video: h264 (High) (avc1 / 0x31637661), yuv420p(tv, bt709, progressive), 2160x3840, 24953 kb/s, 25 fps, 25 tbr, 25 tbn (default)
  Metadata:
    creation_time  : 2021-07-22T11:55:15.000000Z
    handler_name   : L-SMASH Video Handler
    encoder        : AVC coding
Stream mapping:
  Stream #0:0 -> #0:0 (h264 (native) -> h264 (libx264))
Press [q] to stop, [?] for help
[libx264 @ 000002a0d1053d40] using cpu capabilities: MMX2 SSE2Fast SSSE3 SSE4.2 AVX FMA3 BMI2 AVX2 AVX512
[libx264 @ 000002a0d1053d40] profile High, level 5.1, 4:2:0, 8-bit
[libx264 @ 000002a0d1053d40] 264 - core 165 r3223 0480c0 - H.264/MPEG-4 AVC codec - Copyleft 2003-2025 - http://www.videolan.org/x264.html - options: cabac=1 ref=3 deblock=1:0
:0 analyse=0x3:0x113 me=hex subme=7 psy=1 psy_rd=1.00:0.00 mixed_ref=1 me_range=16 chroma_me=1 trellis=1 8x8dct=1 cqm=0 deadzone=21,11 fast_pskip=1 chroma_qp_offset=-2 threads=
12 lookahead_threads=2 sliced_threads=0 nr=0 decimate=1 interlaced=0 bluray_compat=0 constrained_intra=0 bframes=3 b_pyramid=2 b_adapt=1 b_bias=0 direct=1 weightb=1 open_gop=0
weightp=2 keyint=250 keyint_min=25 scenecut=40 intra_refresh=0 rc_lookahead=40 rc=crf mbtree=1 crf=23.0 qcomp=0.60 qpmin=0 qpmx=69 qstep=4 ip_ratio=1.40 aq=1:1.00
Output #0, matroska, to 'Hasil\video.mkv':
  Metadata:
    major_brand   : mp42
    minor_version : 0
    compatible_brands: mp42mp4isomavc1
    creation_time  : Lavf63.1.102
  Stream #0:0(und): Video: h264 (H264 / 0x34363248), yuv420p(tv, bt709, progressive), 2160x3840, q-2-31, 25 fps, 1k tbn (default)
  Metadata:
    encoder      : Lavc63.1.102 libx264
    creation_time : 2021-07-22T11:55:15.000000Z
    handler_name  : L-SMASH Video Handler
  Side data:
    CPB properties: bitrate max/min/avg: 0/0/0 buffer size: 0 vbv_delay: N/A
[frame= 69 fps=4.0 q=29.0 size= 5120KiB time=00:00:02.68 bitrate=15650.4kbits/s speed=0.186x elapsed=0:00:14.37]

```

3. Informasi ke 5 file sesudah dikonversi

```

PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner Hasil/audio.mp3
Input #0, mp3, from 'Hasil/audio.mp3':
  Metadata:
    encoder      : Lavf63.1.102
  Duration: 00:02:27.89, start: 0.025057, bitrate: 128 kb/s
  Stream #0:0: Audio: mp3 (mp3float), 44100 Hz, stereo, fltp, 128 kb/s, start 0.025057
  Metadata:
    encoder      : Lavc63.1.
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner Hasil/images.jpeg
Input #0, image2, from 'Hasil/images.jpeg':
  Duration: 00:00:00.04, start: 0.000000, bitrate: 3968 kb/s
  Stream #0:0: Video: mjpeg (Baseline), yuvj444p(pc, bt470bg/unknown/unknown), 554x554 [SAR 1:1 DAR 1:1], 25 fps, 25 tbr, 25 tbn
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner Hasil/lagu.wav
Input #0, wav, from 'Hasil/lagu.wav':
  Metadata:
    artist       : yung kai
    comment      : https://www.youtube.com/watch?v=98zHKN-xSHk
    date         : 2024
    title        : yung kai - blue (Official Audio)
    encoder      : Lavf63.1.102
  Duration: 00:03:34.36, bitrate: 1536 kb/s
  Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / 0x0001), 48000 Hz, 2 channels, s16, 1536 kb/s
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner Hasil/screenshot.jpg
Input #0, image2, from 'Hasil/screenshot.jpg':
  Duration: 00:00:00.04, start: 0.000000, bitrate: 24853 kb/s
  Stream #0:0: Video: mjpeg (Baseline), yuvj444p(pc, bt470bg/unknown/unknown), 1919x1070 [SAR 1:1 DAR 1919:1070], 25 fps, 25 tbr, 25 tbn
PS C:\Kuliah\Semester 5\Praktikum Sistem Multimedia\Pertemuan01> ffprobe -hide_banner Hasil/video.mkv
Input #0, matroska,webm, from 'Hasil/video.mkv':
  Metadata:
    COMPATIBLE_BRANDS: mp42mp41isomavc1
    MAJOR_BRAND       : mp42
    MINOR_VERSION     : 0
    ENCODER           : Lavf63.1.102
  Duration: 00:00:22.24, start: 0.000000, bitrate: 18650 kb/s
  Stream #0:0: Video: h264 (High), yuv420p(tv, bt709, progressive), 2160x3840, 25 fps, 25 tbr, 1k tbn (default)
  Metadata:
    ENCODER           : Lavc63.1.102 libx264
    HANDLER_NAME      : L-SMASH Video Handler
    DURATION          : 00:00:22.240000000
  
```

4. Analisis Hasil (Tabel)

File	Sebelum Konversi (Folder DokSli)	Sesudah Konversi (Folder Hasil)	Analisis Perubahan
audio.wav → audio.mp3	Format: WAV (Uncompressed PCM) Bitrate: 1411 kb/s Sample Rate: 44.1 kHz	Format: MP3 (Compressed Lossy) Bitrate: 128 kb/s Sample Rate: 44.1 kHz	Terjadi kompresi lossy dari WAV ke MP3. Bitrate berkurang drastis dari 1411 kb/s menjadi 128 kb/s, yang membuat ukuran file jauh lebih kecil.
lagu.mp3 → lagu.wav	Format: MP3 Bitrate: 320 kb/s (Audio) / 366 kb/s	Format: WAV Bitrate:	File diekstrak ke format uncompressed (WAV). Bitrate audio meningkat

File	Sebelum Konversi (Folder DokSli)	Sesudah Konversi (Folder Hasil)	Analisis Perubahan
	(Total) Sample Rate: 48 kHz Fitur: Memiliki <i>Attached Pic</i> (Cover Art PNG 1280x720)	1536 kb/s Sample Rate: 48 kHz Fitur: Sampul gambar terlepas	menjadi 1536 kb/s, namun metadata sampul album (<i>attached pic</i>) dihilangkan.
video.mp4 → video.mkv	Container: MP4 Codec: H.264 / AVC Bitrate: 24.956 kb/s (~24.9 Mbps) Resolusi: 2160x3840 (4K)	Container: MKV (Matroska) Codec: H.264 (re-encoded via libx264) Bitrate: 18.650 kb/s (~18.6 Mbps) Resolusi: 2160x3840 (4K)	Format kontainer berubah dari MP4 ke MKV. Karena dilakukan proses <i>re-encoding</i> default (CRF 23), bitrate video berkurang dari 24.9 Mbps menjadi 18.6 Mbps tanpa mengubah resolusi.
screeshoot.png → screeshoot.jpg	Format: PNG (Lossless) Color Space: RGBA (Memiliki kanal transparansi/Alpha) Bitrate: N/A	Format: JPEG (Lossy / MJPEG) Color Space: YUV444P Bitrate: 24.853 kb/s	Konversi dari PNG ke JPG menghilangkan kanal transparansi (Alpha channel) dan menerapkan kompresi lossy JPEG.
images.jpg → images.jpeg	Format: JPEG (.jpg)	Format: JPEG (.jpeg) Bitrate:	Hanya perubahan ekstensi dan konversi ulang ringan antar-

File	Sebelum Konversi (Folder DokSli)	Sesudah Konversi (Folder Hasil)	Analisis Perubahan
	Bitrate: 4160 kb/s Resolusi: 554x554	3968 kb/s Resolusi: 554x554	format JPEG. Bitrate mengalami penurunan kecil dari 4160 kb/s ke 3968 kb/s.

1.5. Kesimpulan

Proses konversi multimedia menggunakan FFmpeg pada praktikum ini menunjukkan bagaimana perubahan format, kontainer, dan skema enkoding memengaruhi karakteristik data multimedia:

- Efisiensi Kompresi dan Ukuran File: Konversi dari format uncompressed atau lossless (seperti WAV dan PNG) ke format lossy (seperti MP3 dan JPG) berhasil memotong bitrate secara signifikan. Hal ini berdampak langsung pada pengecilan ukuran file sehingga lebih efisien untuk penyimpanan maupun transmisi data.
- Kualitas vs. Bitrate (Kasus Re-Encoding): Mengubah format lossy ke uncompressed (seperti MP3 ke WAV) memang meningkatkan bitrate file secara teknis, namun tidak dapat mengembalikan kualitas audio yang sudah hilang saat kompresi awal. Pada pemrosesan video (MP4 ke MKV), re-encoding default menggunakan codec libx264 mampu menekan bitrate dari 24.9 Mbps menjadi 18.6 Mbps tanpa mengubah resolusi asli (4K).
- Perubahan Fitur dan Metadata: Proses transkoding juga memengaruhi struktur file, seperti hilangnya kanal transparansi (Alpha channel) saat mengubah PNG ke JPG, serta terlepasnya metadata attached pic (cover album) saat file MP3 dikonversi menjadi WAV standar.

LAMPIRAN

Instalasi software etc.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\MyBook Hype> python --version
Python 3.12.12
PS C:\Users\MyBook Hype> pip --version
pip 26.2.1 from C:\Users\MyBook Hype\AppData\Local\Programs\Python\Python311\Lib\site-packages\pip (python 3.11)
PS C:\Users\MyBook Hype>
```

Figure 1 Bukti Istalasi Python dan Pip

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\MyBook Hype> pip install pillow numpy scipy opencv-python scikit-image PyWavelets jupyter
Requirement already satisfied: pillow in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (12.3.0)
Requirement already satisfied: numpy in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.4.0)
Requirement already satisfied: scipy in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (1.17.1)
Requirement already satisfied: opencv-python in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (5.0.9.92)
Requirement already satisfied: scikit-image in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (0.26.0)
Requirement already satisfied: PyWavelets in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (1.9.0)
Collecting jupyter
  Using cached jupyter-1.1.1-py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: networkx<3.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from scikit-image) (3.6.1)
Requirement already satisfied: imageio<2.35.0, >=2.23 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from scikit-image) (2.37.4)
Requirement already satisfied: tifffile<2023.8.12 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from scikit-image) (2026.3.3)
Requirement already satisfied: packaging<=21 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from scikit-image) (26.3)
Requirement already satisfied: lazy-loader<=0.4 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from scikit-image) (0.6)
Collecting notebook (from jupyter)
  Using cached notebook-7.6.3-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: jupyter-console in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter) (6.6.3)
Requirement already satisfied: nbconvert in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter) (7.17.1)
Requirement already satisfied: ipykernel in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter) (7.3.0)
Requirement already satisfied: ipynbdatasets in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter) (0.3.2)
Collecting jupyterlab (from jupyter)
  Using cached jupyterlab-4.6.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: comm<0.1.1 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (0.2.3)
Requirement already satisfied: debugpy<1.6.5 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (1.8.22)
Requirement already satisfied: ipython<7.23.1 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (9.17.1)
Requirement already satisfied: jupyter-client<8.0, >=8.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (8.10.0)
Requirement already satisfied: jupyter-core<10.0, >=8.1 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (5.9.1)
Requirement already satisfied: matplotlib-inline<=0.1 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (0.2.2)
Requirement already satisfied: nest-asyncio<=1.7.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (1.7.1)
Requirement already satisfied: psutil<=5.7 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (7.2.2)
Requirement already satisfied: pyzmq<25 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (27.2.0)
Requirement already satisfied: tornado<=6.4 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (6.5.10)
Requirement already satisfied: traitlets<5.4.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipykernel->jupyter) (5.16.1)
Requirement already satisfied: colorama<=0.4.4 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (0.4.4)
Requirement already satisfied: ipython-pygments-lexer<=1.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (1.1.1)
Requirement already satisfied: jedi<=0.18.2 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (0.20.0)
Requirement already satisfied: pygments<=2.19.0, >=2.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (3.0.53)
Requirement already satisfied: stack_data<=0.6.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (0.2.0)
Requirement already satisfied: typing_extensions<=4.6 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipython<7.23.1->ipykernel->jupyter) (4.16.0)
Requirement already satisfied: wcwidth<=0.1.1 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from prompt_toolkit3.1.0, >=3.0.41->ipython<7.23.1->ipykernel->jupyter) (0.9.1)
Requirement already satisfied: parso<0.9.0, >=0.8.6 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jedi<=0.18.2->ipython<7.23.1->ipykernel->jupyter) (0.8.7)
Requirement already satisfied: python-dateutil<=2.9 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter-client<8.0, >=8.0->ipykernel->jupyter) (2.0.0.post0)
Requirement already satisfied: platformdirs<=2.5 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jupyter-core<10.0, >=8.1->ipykernel->jupyter) (4.12.0)
Requirement already satisfied: six<1.5 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from python-dateutil<=2.9->ipython<7.23.1->ipykernel->jupyter) (1.17.0)
Requirement already satisfied: executing<1.2.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from stack_data<=0.6.0->ipython<7.23.1->ipykernel->jupyter) (2.2.1)
Requirement already satisfied: asttokens<=2.1.0 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from stack_data<=0.6.0->ipython<7.23.1->ipykernel->jupyter) (3.0.2)
Requirement already satisfied: pure-eval in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from stack_data<=0.6.0->ipython<7.23.1->ipykernel->jupyter) (0.2.4)
Requirement already satisfied: ipynbdatasets<=0.3.16 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ipynbdatasets->jupyter) (0.3.16)
```

Figure 2 Bukti Instalasi Pustaka Praktikum

```
PS C:\Users\MyBook Hype> ffmpeg -version
ffmpeg version 9.0.2-essentials_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --
--enable-fontconfig --enable-iconv --enable-gnutls --enable-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zl
ib --enable-librt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-libx264 --e
nable-libx265 --enable-libxvid --enable-libaom --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-li
bass --enable-libfreetype --enable-libfribidi --enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg
--enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --enable-ffnvcodec --en
able-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-opengl --enable-libgme --enable-libopenmpt --enable-li
bopencore-amrwb --enable-libmp3lame --enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrnb
--enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
libavutil 61. 1.102 / 61. 1.102
libavcodec 63. 1.102 / 63. 1.102
libavformat 63. 1.102 / 63. 1.102
libavdevice 63. 1.102 / 63. 1.102
libavfilter 12. 1.102 / 12. 1.102
libswscale 10. 1.102 / 10. 1.102
libswresample 7. 1.102 / 7. 1.102
```

Figure 3 Bukti Penambahan Environment Variabel FFMPEG

```

Exiting with exit code 0
PS C:\Users\MyBook Hype> ffprobe -version
ffprobe version 9.0.2-essentials_build-www.gyan.dev Copyright (c) 2007-2026 the FFmpeg developers
built with gcc 16.2.0 (Rev3, Built by MSYS2 project)
configuration: --enable-gpl --enable-version3 --enable-static --disable-w32threads --disable-autodetect --enable-cairo --
--enable-fontconfig --enable-iconv --enable-gnutls --enable-libxml2 --enable-gmp --enable-bzlib --enable-lzma --enable-zl
ib --enable-libsrt --enable-libssh --enable-libzmq --enable-avisynth --enable-sdl2 --enable-libwebp --enable-libx264 --e
nable-libx265 --enable-libxvid --enable-libaom --enable-libopenjpeg --enable-libvpx --enable-mediafoundation --enable-li
bass --enable-libfreetype --enable-libfribidi --enable-libharfbuzz --enable-libvidstab --enable-libvmaf --enable-libzimg
--enable-amf --enable-cuda-llvm --enable-cuvid --enable-dxva2 --enable-d3d11va --enable-d3d12va --enable-ffnvcodec --en
able-libvpl --enable-nvdec --enable-nvenc --enable-vaapi --enable-openal --enable-libgme --enable-libopenmpt --enable-li
bopencore-amrwb --enable-libmp3lame --enable-libtheora --enable-libvo-amrwbenc --enable-libgsm --enable-libopencore-amrn
b --enable-libopus --enable-libspeex --enable-libvorbis --enable-librubberband
libavutil      61. 1.102 / 61. 1.102
libavcodec     63. 1.102 / 63. 1.102
libavformat    63. 1.102 / 63. 1.102
libavdevice    63. 1.102 / 63. 1.102
libavfilter    12. 1.102 / 12. 1.102
libswscale     10. 1.102 / 10. 1.102
libswresample  7. 1.102 / 7. 1.102

```

Figure 4 Bukti Penambahan Environment Variabel FFBROBE

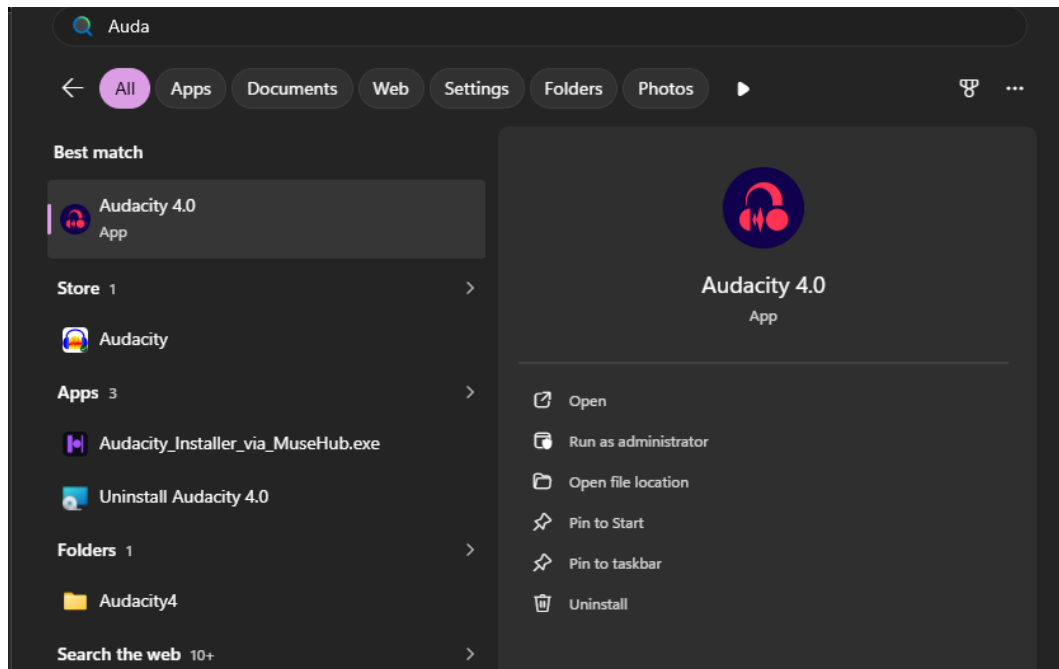


Figure 5 Bukti Instalasi Audacity 4.0

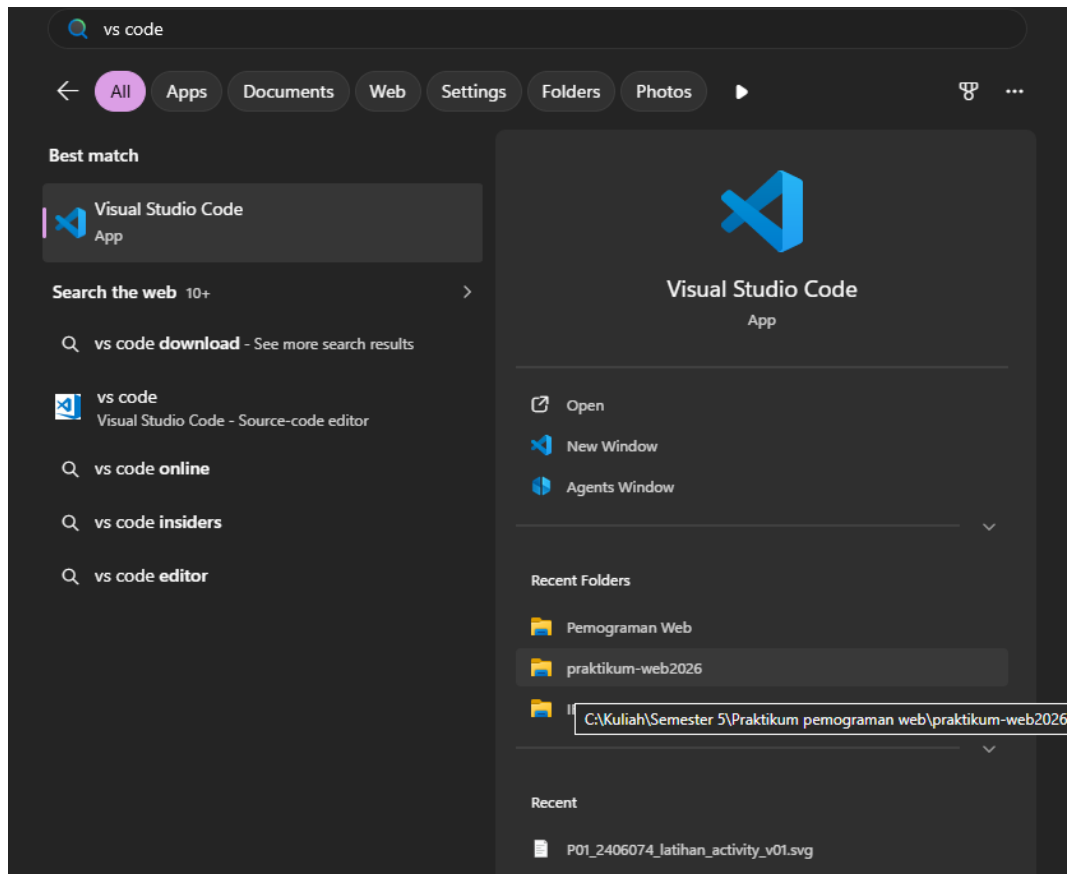


Figure 6 Bukti Instalasi Visualcode Studio